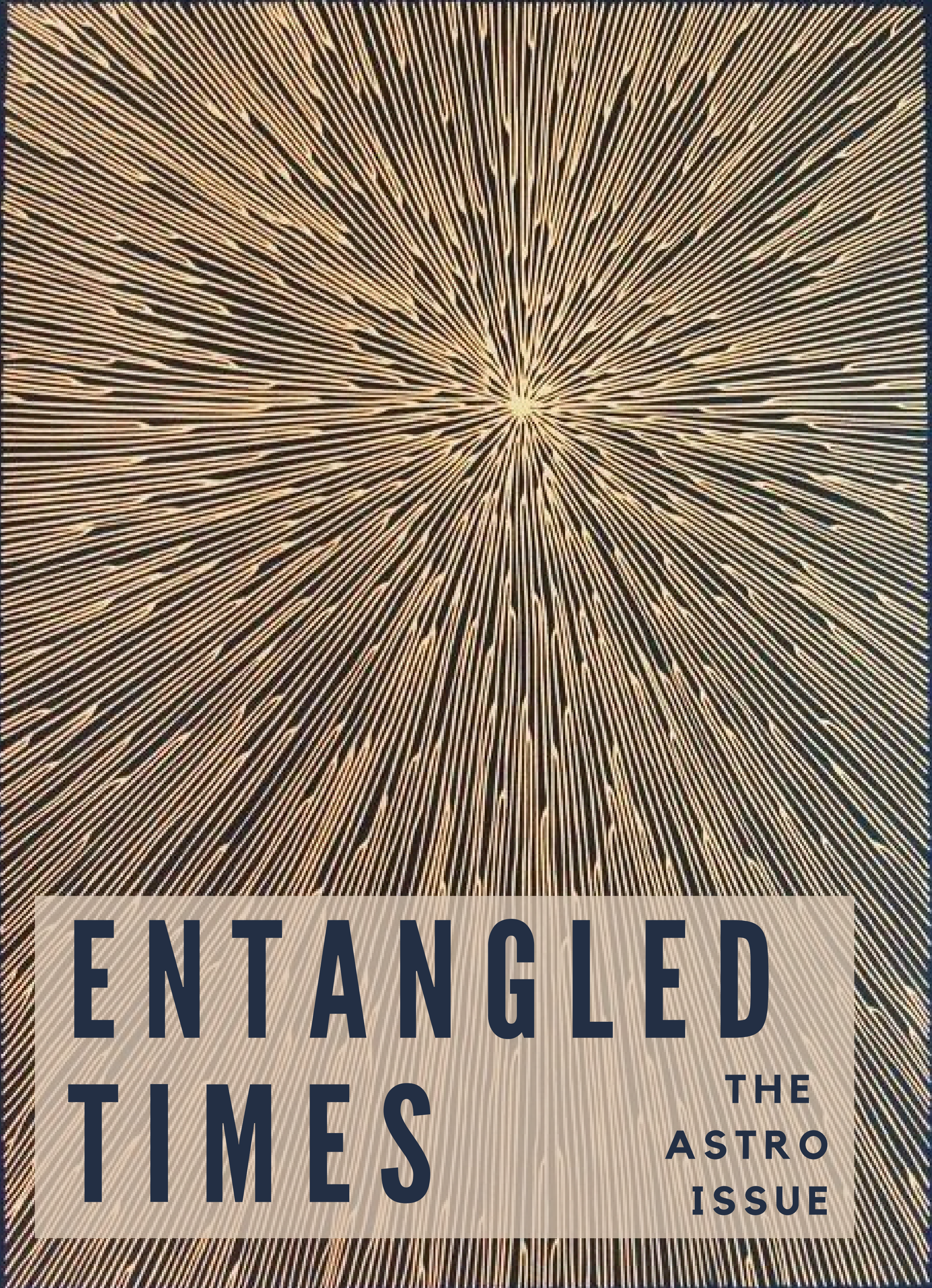




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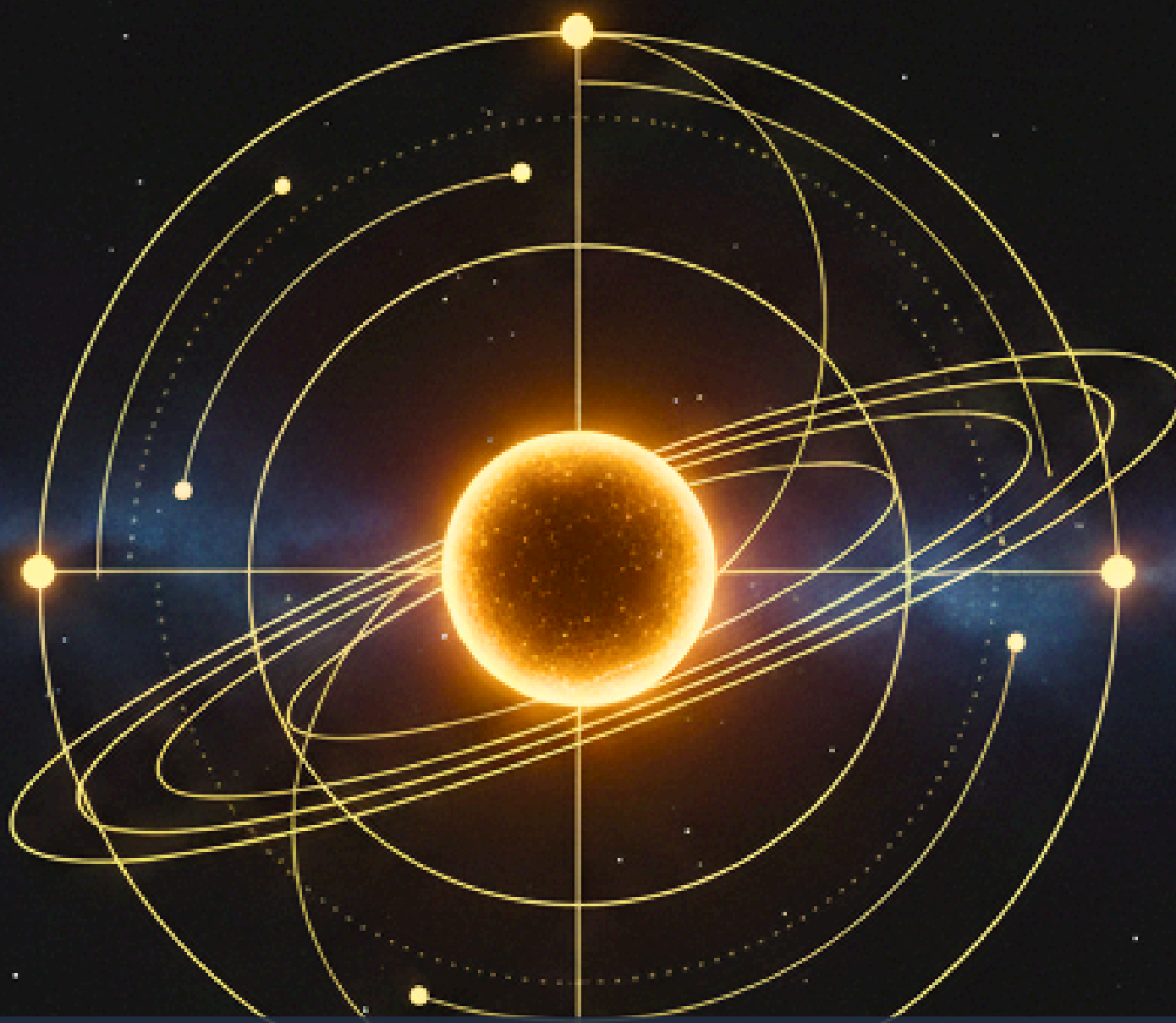
THE
ASTRO
ISSUE

TABLE OF CONTENTS



1. WHAT IS
ASTROPHYSICS?
2. THE VAST UNIVERSE
3. STARS
4. GRAVITY
5. BLACK HOLES
6. LIGHT
7. HOW WE USE
ASTROPHYSICS?

WHAT IS ASTROPHYSICS?



ASTROPHYSICS IS THE BRANCH OF SCIENCE THAT STUDIES STARS, PLANETS, GALAXIES, AND THE UNIVERSE. IT COMBINES PHYSICS AND ASTRONOMY TO UNDERSTAND HOW SPACE WORKS AND HOW THE UNIVERSE BEGAN.

WHEN YOU LOOK UP AT THE NIGHT SKY AND SEE STARS, YOU ARE ACTUALLY LOOKING AT OBJECTS THAT ARE MILLIONS OR EVEN BILLIONS OF KILOMETERS AWAY. ASTROPHYSICS HELPS US ANSWER BIG QUESTIONS LIKE:

- HOW ARE STARS BORN?**
- WHAT ARE BLACK HOLES?**
- HOW DID THE UNIVERSE BEGIN?**
- ARE WE ALONE IN THE UNIVERSE?**

THE VAST UNIVERSE



THE UNIVERSE IS INCREDIBLY
LARGE. IT CONTAINS:

- PLANETS LIKE EARTH
- STARS LIKE OUR SUN
- GALAXIES MADE UP OF
BILLIONS OF STARS
- BLACK HOLES, NEBULAE, AND
MORE

OUR HOME IS IN THE MILKY WAY
GALAXY, WHICH CONTAINS
HUNDREDS OF BILLIONS OF STARS.
AND THE MILKY WAY IS JUST ONE
OF BILLIONS OF GALAXIES IN THE
UNIVERSE.

STARS: COSMIC POWERHOUSES

STARS ARE GIANT BALLS OF HOT GAS THAT PRODUCE LIGHT AND HEAT. THEY SHINE BECAUSE OF A PROCESS CALLED NUCLEAR FUSION.

IN THE CENTER OF A STAR:

- HYDROGEN ATOMS COMBINE TO FORM HELIUM
 - THIS RELEASES HUGE AMOUNTS OF ENERGY
- THIS ENERGY IS WHAT MAKES STARS GLOW.

FUN FACT: LIGHT FROM SOME STARS TAKES MILLIONS OF YEARS TO REACH EARTH. THAT MEANS WE SEE THEM AS THEY WERE IN THE PAST.

GRAVITY: THE COSMIC FORCE

GRAVITY IS THE FORCE THAT HOLDS
THE UNIVERSE TOGETHER.

- IT KEEPS PLANETS ORBITING
STARS
- IT KEEPS MOONS ORBITING
PLANETS
- IT HOLDS GALAXIES TOGETHER

THE MORE MASSIVE AN OBJECT IS,
THE STRONGER ITS GRAVITY.
THAT'S WHY:

- THE SUN KEEPS ALL THE
PLANETS IN ORBIT
- EARTH'S GRAVITY KEEPS US ON
THE GROUND

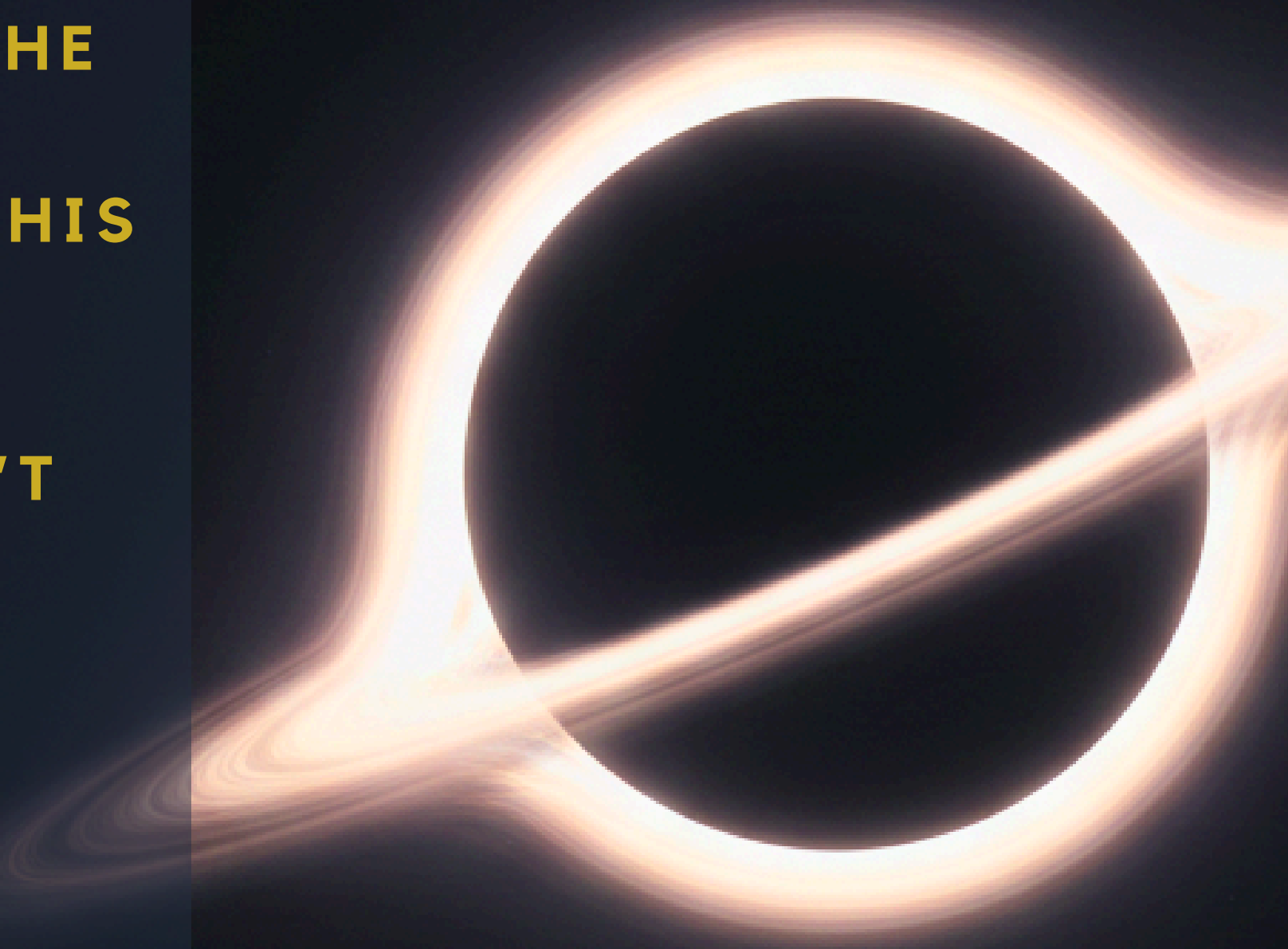
BLACK HOLES: THE COSMIC MYSTERIES

A BLACK HOLE IS A PLACE IN SPACE WHERE GRAVITY IS SO STRONG THAT NOTHING CAN ESCAPE, NOT EVEN LIGHT.

BLACK HOLES FORM WHEN VERY MASSIVE STARS COLLAPSE AT THE END OF THEIR LIVES.

AROUND A BLACK HOLE IS A BOUNDARY CALLED THE EVENT HORIZON. ONCE SOMETHING CROSSES THIS BOUNDARY, IT CANNOT COME BACK.

EVEN THOUGH WE CAN'T SEE BLACK HOLES DIRECTLY, SCIENTISTS DETECT THEM BY OBSERVING HOW THEY AFFECT NEARBY STARS AND GAS.



LIGHT: WINDOW TO THE UNIVERSE

WE LEARN ABOUT SPACE
MOSTLY THROUGH LIGHT.
TELESCOPES COLLECT LIGHT
FROM DISTANT OBJECTS AND
HELP US SEE:

- PLANETS
- STARS
- GALAXIES
- NEBULAE

DIFFERENT TYPES OF LIGHT REVEAL
DIFFERENT INFORMATION:

- VISIBLE LIGHT SHOWS WHAT WE
CAN SEE
- INFRARED SHOWS HEAT
- X-RAYS SHOW EXTREMELY HOT
AND ENERGETIC OBJECTS

THIS IS HOW SCIENTISTS STUDY THE
UNIVERSE WITHOUT LEAVING EARTH.



HOW WE USE ASTROPHYSICS

ASTROPHYSICS HELPS US IN MANY WAYS:

- **SATELLITES: USED FOR GPS, WEATHER, AND COMMUNICATION**
- **SPACE EXPLORATION: ROCKETS AND PROBES STUDY OTHER PLANETS**
- **TECHNOLOGY: MANY INVENTIONS, LIKE CAMERA SENSORS, CAME FROM SPACE RESEARCH**
- **UNDERSTANDING EARTH: STUDYING OTHER PLANETS HELPS US LEARN ABOUT OUR OWN**

ABOUT THE AUTHOR

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**MY NAME IS ISHAAN KAPOOR AND I'M AN
IBDP GRADE 11 STUDENT WITH A PASSION
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